

A Lean companion to Classical Lottery in Action

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May 22, 2026

Abstract

This note documents the LEAN 4 artifact `ClassicalLotteryInAction.lean` accompanying the published Management Science paper *Classical Lottery in Action: Quantifying Risk and Evaluating Uncertainty*. The artifact formalizes the local classical-lottery infrastructure, the average-utility consequences used for classical lotteries, matching-frequency constructions for two-prize acts, and the public wrapper statements for the smooth-ambiguity representation and ambiguity-attitude results. Its main design choice is audit transparency: non-local representation-theoretic ingredients are exposed as named bridge hypotheses, while the public theorem path is checked by LEAN. The axiom audit for the five public results reported here contains only `[propext, Classical.choice, Quot.sound]`; the wrapper hypotheses remain ordinary theorem assumptions, not hidden axioms. A companion spin-out, `MechanizedDecisionTheoryWakkerDK.pdf`, is cited as ongoing work on the Wakker/Debreu-Koopmans representation layer imported by the Management Science artifact.

1 Purpose of the note

The economic and conceptual contribution is the published Management Science article [1]. This short companion note has a narrower purpose: it explains what the `LEANartifact` proves, which theorem-shaped ingredients it deliberately leaves as named bridges, and how to reproduce the artifact-level axiom audit.

The source file is

`Classical Lottery/ClassicalLotteryInAction.lean`.

It formalizes classical lotteries as non-empty lists, acts as state-indexed lotteries, and preferences as binary relations on acts. The file then develops the operations and predicates used in the paper: concatenation, replication, deletion of matched sub-lotteries, relative frequencies, average utility, two-prize matching frequencies, smooth representations, and ambiguity-attitude predicates.

The artifact is not a monolithic formal proof of every representation theorem invoked in the paper. Instead, it follows a named-bridge architecture: local algebraic, order-theoretic, and finite-sum facts are proved directly, while monograph-scale or non-local representation steps are imported through explicit `Prop`-valued hypotheses. This makes the formal boundary inspectable.

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2 Artifact surface

The public-facing Lean statements tracked by this note are as follows.

- `prop_average_utility`: Proposition 1, the average-utility characterization for classical lotteries. The easy direction from representation to Cancellation and Archimedeanity is proved; the representation-theorem direction is the named bridge `AverageUtilityHardDirection`.
- `lem_gap_filling`: the two-prize matching-frequency gap-filling lemma. Its reverse implication uses the named strict-separation bridge `MatchingFrequencyStrictSeparation`.
- `thm_smooth_model`: the smooth-representation theorem under explicit Wakker/Debreu–Koopmans-style sufficiency and regularity wrappers.
- `matching_freq_smooth_formula`: the normalized two-prize formula for behavioral matching frequency, with the inverse- ψ supremum step named as `MatchingFrequencySmoothFormulaBridge`.
- `prop_aversion_or_seeking`: the ambiguity-aversion/ambiguity-seeking curvature statement. The easy directions from curvature to behavior are proved; behavioral recovery of curvature is isolated as `AmbiguityAttitudeCurvatureBridge`.

The source also defines Axiom 8, `Solvability`, and proves boundary lemmas about solvability, but the five rows above are the main theorem-level audit targets corresponding to the formalization-status paragraph in the published paper.

3 Named-bridge interface

Table 1 records the bridge interface exposed by `ClassicalLotteryInAction.lean`. The important point is that these are Lean assumptions at theorem boundaries, not additional Lean axioms.

The matching-frequency bridge has an additional useful reduction inside the artifact. `TwoPrizeStrictRelFreq` is theorem-backed once `AverageUtilityHardDirection` supplies an average-utility representation, and `TwoPrizeDenseness` is derived from primitive `Denseness` and `Monotonicity` under the same hard-direction bridge. Thus the `_of_averageUtilityHardDirection` route records a single named path from the Proposition 1 representation spine to the gap-filling strict-separation hypothesis.

4 Axiom audit

The source file contains explicit `#print axioms` commands for the five public theorem targets. Running

```
lake env lean "Classical Lottery\ClassicalLotteryInAction .lean"
```

prints the audit in Table 2. The result is intentionally uniform: every row depends only on the standard Lean/mathlib foundations [`propext`, `Classical.choice`, `Quot.sound`].

This audit should be read together with the named-bridge table. The statement “no extra axioms” does not mean that the Wakker/Debreu–Koopmans representation theorem, strict two-prize separation, inverse- ψ matching-frequency identification, or curvature-recovery arguments have been magically proved. It means that the artifact exposes those obligations explicitly as parameters, and the checked code downstream of those parameters introduces no additional global assumptions.

5 Relation to the Wakker/Debreu–Koopmans spin-out

The file `MechanizedDecisionTheoryWakkerDK.pdf` is a separate work-in-progress note on the Wakker additive-representation and Debreu–Koopmans concavity layer [2]. Its certificate vocabulary is complementary to the bridge surface in `ClassicalLotteryInAction.lean`: it studies construction certificates such as `hConstruct`, `hglobal`, `haff`, `hConc`, and `hPairConc`, together with a certificate checklist in `WakkerDebreuKoopmans.lean`. That spin-out is the appropriate home for the multi-month theorem-prover work behind the imported Wakker/Debreu–Koopmans layer.

The Management Science artifact therefore has a stable publication role: it states and checks the classical-lottery reduction, matching-frequency infrastructure, and smooth-model theorem boundaries used by the published paper. The Wakker/Debreu–Koopmans spin-out has a different role: to progressively discharge the imported representation certificates without changing the public interface consumed here.

6 Reproducibility notes

The audit command used for this note was run from the repository root on Windows:

```
lake env lean "Classical Lottery\ClassicalLotteryInAction .lean".
```

The relevant output was:

```
'ClassicalLottery.Preference.prop_average_utility' depends on axioms:
[proptext, Classical.choice, Quot.sound]
'ClassicalLottery.Preference.lem_gap_filling' depends on axioms:  [proptext,
Classical.choice, Quot.sound]
'ClassicalLottery.Preference.thm_smooth_model' depends on axioms:  [proptext,
Classical.choice, Quot.sound]
'ClassicalLottery.Preference.matching_freq_smooth_formula' depends on axioms:
[proptext, Classical.choice, Quot.sound]
'ClassicalLottery.Preference.prop_aversion_or_seeking' depends on axioms:
[proptext, Classical.choice, Quot.sound].
```

References

- [1] Jingyuan Li, Ilia Tsetlin, and Fan Wang. *Classical Lottery in Action: Quantifying Risk and Evaluating Uncertainty*. Management Science, 2026.
- [2] Jingyuan Li, Ilia Tsetlin, and Fan Wang. *Certificate-Driven Mechanization of Wakker–Debreu–Koopmans Additive Representation in Lean 4*. Source draft `MechanizedDecisionTheoryWakkerDK.pdf`, ongoing work, 2026.
- [3] Peter P. Wakker. *Additive Representations of Preferences: A New Foundation of Decision Analysis*. Kluwer Academic Publishers, 1989.
- [4] Gerard Debreu and Tjalling C. Koopmans. Additively decomposed quasiconvex functions. *Mathematical Programming and Related Techniques*, 1960.

Bridge	Lean role	Mathematical content named by the bridge
AverageUtility HardDirection P	Input to <code>prop_average_utility</code>	The vNM/Wakker-style hard direction: Cancellation plus Archimedeanity produce an average-utility representation on constant classical lotteries.
MatchingFrequency StrictSeparation P	Input to <code>lem_gap_filling</code>	On two-prize acts, strict preference implies strict separation of matching frequencies. The file further decomposes it into <code>TwoPrizeDenseness</code> and <code>TwoPrizeStrictRelFreq</code> .
SmoothRepresentation Sufficiency Pref	Forward direction of <code>thm_smooth_model</code>	From non-triviality and Axioms 1–7, obtain a smooth representation. This is the Wakker/Debreu–Koopmans representation-and-patching import.
SmoothRepresentation Regularity Pref	Reverse regularity part of <code>thm_smooth_model</code>	From a smooth representation, recover Denseness, Continuity, and Consistent Aggregation; Weak Order, Cancellation, Archimedeanity, and Monotonicity are derived locally.
MatchingFrequency SmoothFormulaBridge Pref R	Input to <code>matching_freq_smooth_formula</code>	Under $u(x) = 1, u(y) = 0$, identify the behavioral matching frequency with $\psi^{-1}(\int \psi(r_x(f(s))) dP)$. The integrand simplification is proved separately.
AmbiguityAttitude CurvatureBridge Pref R	Input to <code>prop_aversion_or_seeking</code>	Converse Debreu–Koopmans-style curvature recovery: ambiguity aversion implies concavity of ψ , and ambiguity seeking implies convexity.

Table 1: Named bridges in `ClassicalLotteryInAction.lean`.

Public theorem	Paper role	Axiom dependencies reported by Lean
<code>prop_average_utility</code>	Proposition 1: average utility for classical lotteries	<code>[propext,</code> <code>Classical.choice,</code> <code>Quot.sound]</code>
<code>lem_gap_filling</code>	Lemma 1: matching-frequency gap filling on two-prize acts	<code>[propext,</code> <code>Classical.choice,</code> <code>Quot.sound]</code>
<code>thm_smooth_model</code>	Theorem 1: smooth-representation wrapper theorem	<code>[propext,</code> <code>Classical.choice,</code> <code>Quot.sound]</code>
<code>matching_freq_smooth_formula</code>	Equation/formula for normalized smooth matching frequency	<code>[propext,</code> <code>Classical.choice,</code> <code>Quot.sound]</code>
<code>prop_aversion_or_seeking</code>	Proposition 2: ambiguity attitudes and curvature	<code>[propext,</code> <code>Classical.choice,</code> <code>Quot.sound]</code>

Table 2: Axiom audit from the artifact. The bridge hypotheses in Section 3 are theorem assumptions and therefore do not appear as global Lean axioms.